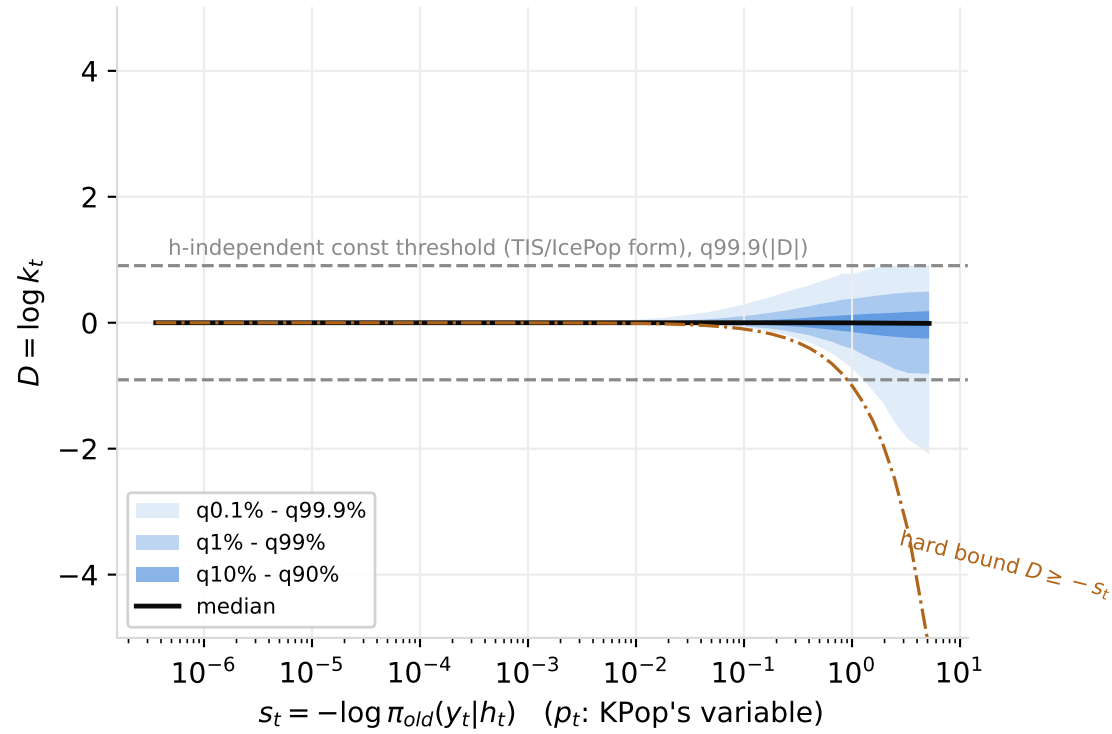
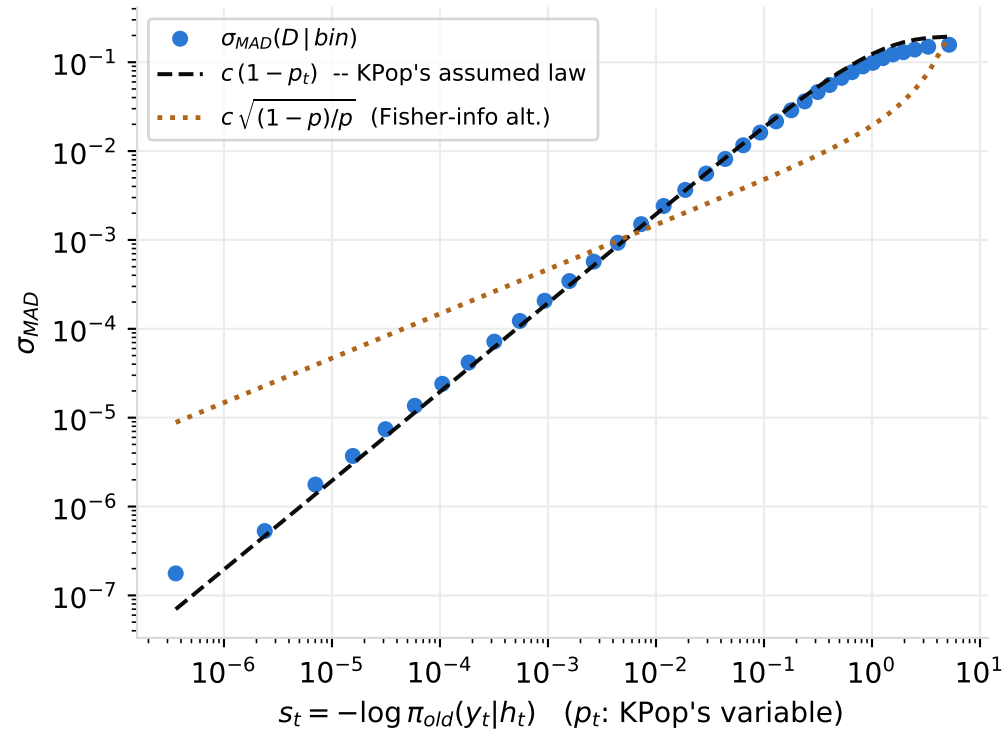


V8 -- empirical law of $\log k_t$ vs the assumed laws of TIS / IcePop / KPop (moe, 12,293,115 tokens, axis: ptrain)

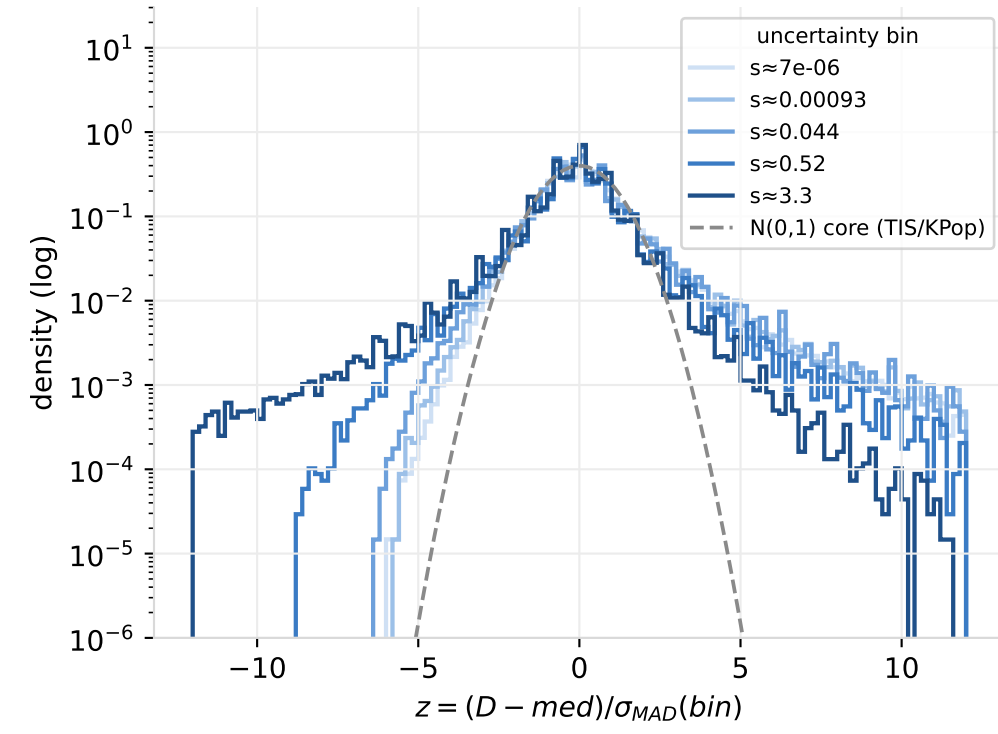
(a) conditional quantile fan of $\log k_t | s$



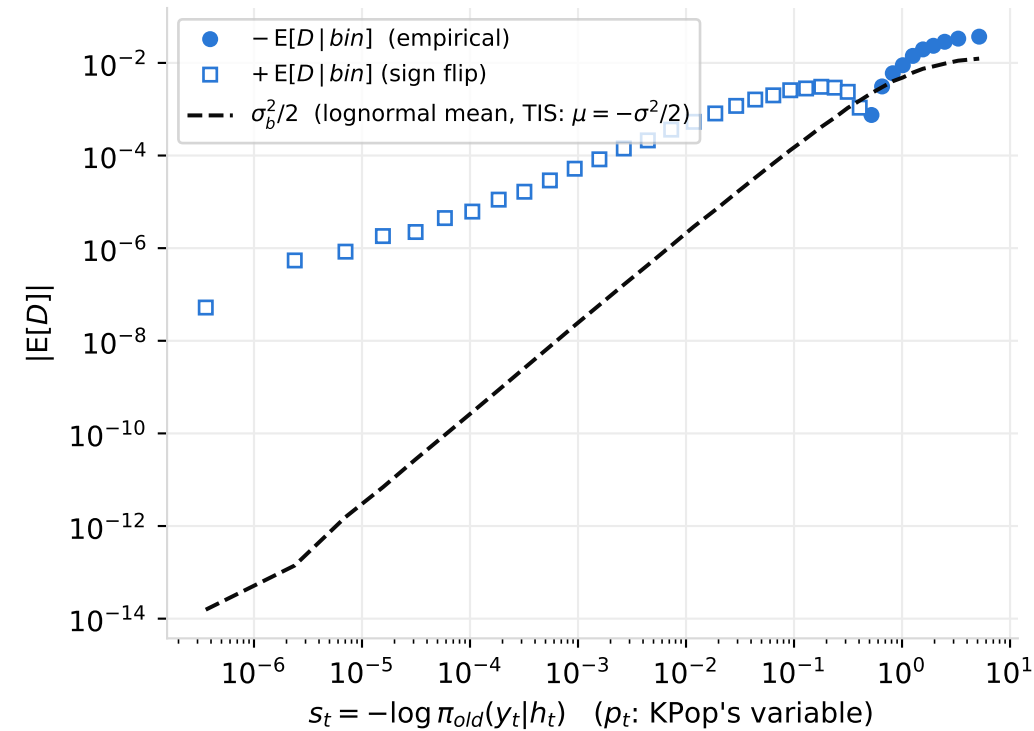
(b) scale law: TIS/IcePop assume flat; KPop assumes $\propto (1 - p_t)$



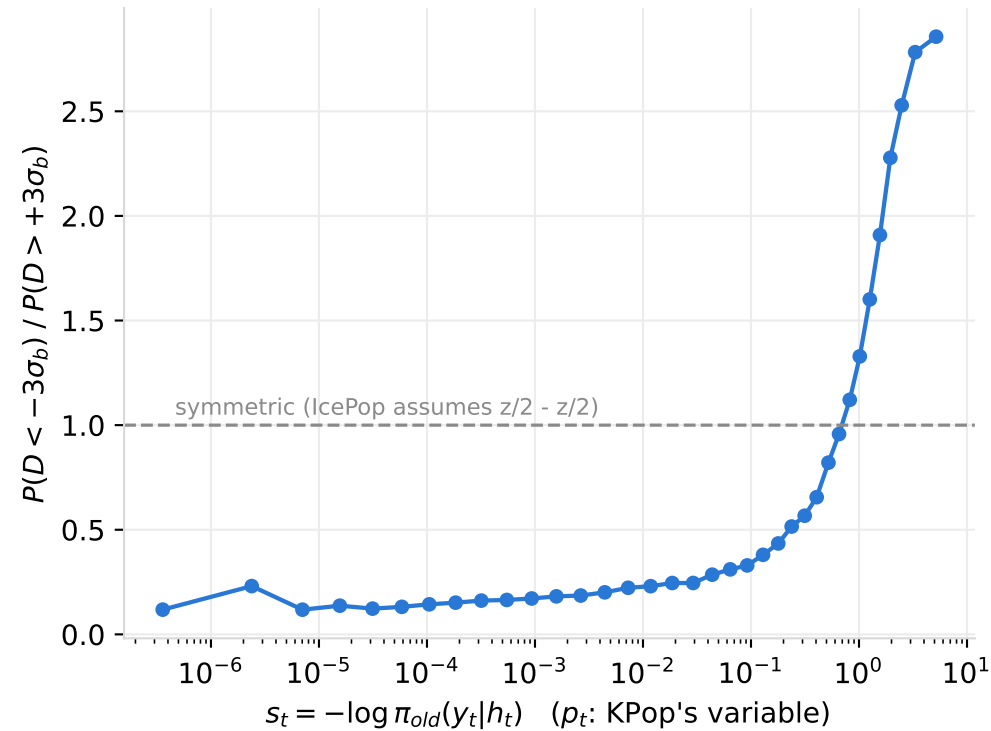
(c) shape: near scale-family, non-Gaussian tails



(d) mean: KPop assumes 0; TIS assumes $-\sigma^2/2$



(e) outlier-branch asymmetry vs IcePop's symmetric mixture



(f) IcePop Exp(λ) branch: straight iff exponential

